

Problem Statement - Electric Vehicle Charging Optimization Agent

The Challenge

With the rapid growth of electric vehicles (EVs), charging stations often face high demand, uneven usage, and inefficient energy distribution. Peak-hour charging can overload the grid and increase electricity costs for both operators and users. There is a need for an intelligent system that can analyze charging patterns and optimize EV charging schedules.

The Objective

Build an **Electric Vehicle Charging Optimization Agent** that analyzes charging station usage data and provides intelligent recommendations to optimize charging schedules and energy usage.

1. **Charging Pattern Analysis** – Analyze historical EV charging data to understand peak usage hours, charging demand, and station utilization.
2. **Charging Schedule Optimization** – Recommend optimal charging schedules to balance demand and reduce peak-hour congestion.
3. **Energy Demand Prediction** – Predict future charging demand based on historical patterns and user behavior.
4. **Cost Optimization Insights** – Suggest charging during low-tariff periods to reduce electricity costs for EV users and charging station operators.

Tech Stack (Mandatory)

- IBM Langflow / IBM Orchestrate + IBM Models